

Supplementary Material for DeGro: Target Determinacy Verification and Grounded Repair for Executable Mathematical Reasoning

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This document provides proof details, evaluation protocol, selected extended results, and reproducibility identifiers. The main paper is self-contained.

A Formal Details

Let a structured formalization be $F = (V, D, C, q)$, where V is a finite set of typed variables, D contains domain restrictions, C contains translated source constraints, and $q(V)$ is the target. Write

$$\text{Sol}(F) = \{m \mid m \models D \wedge C\}.$$

The target is determinate when all feasible assignments agree:

$$\text{Det}(F, q) \iff \forall m_1, m_2 \in \text{Sol}(F), q(m_1) = q(m_2). \quad (1)$$

For two disjoint renamings $V^{(1)}$ and $V^{(2)}$, define

$$\Phi(F, q) = F^{(1)} \wedge F^{(2)} \wedge q(V^{(1)}) \neq q(V^{(2)}). \quad (2)$$

Witness equivalence. Assuming semantics-preserving compilation, $\Phi(F, q)$ is satisfiable exactly when two feasible assignments disagree on q . A satisfying assignment projects to the two witnesses; conversely, any such pair satisfies Eq. (2). Hence, if F is satisfiable and $\Phi(F, q)$ is unsatisfiable, Eq. (1) holds. The initial satisfiability check is necessary because an inconsistent F would otherwise yield a vacuous certificate.

Repair gate. For candidate constraint c and a frozen set $\mathcal{R}$ of permitted domain-semantic rules,

$$\begin{aligned} \text{Accept}(c) \iff & \text{Compiles}(c) \wedge \text{Grounded}(c, P, \mathcal{R}) \\ & \wedge \text{SAT}(F \wedge c) \wedge \neg(F \models c) \wedge \text{Det}(F \wedge c, q). \end{aligned} \quad (3)$$

Thus every accepted edit is supported, consistent, novel, and target-sufficient. If neither of two proposals passes, the original F is preserved. Exact-span provenance is an auditable proxy, not a proof of semantic equivalence.

B Reference Algorithm and Trusted Boundary

The bounded controller performs:

1. parse one typed `MODEL SPEC` and reject unsupported expressions;
2. check $\text{SAT}(F)$, then $\Phi(F, q)$;
3. if ambiguous, record two target-divergent witnesses;
4. request one grounded constraint or `ABSTAIN`;
5. apply Eq. (3), stopping after at most two proposals.

The restricted parser, typed compiler, Z3, and deterministic gate form the trusted boundary; the language model does not. The parser rejects undeclared identifiers, arbitrary calls, attribute access, and unsupported operators. Both solving and verification consume the same normalized specification. Logs retain initial and final states, witnesses, proposals, cited spans, rejection reasons, model configuration, and prompt hashes.

C Controlled Paired Evaluation

For a validated gold specification $F^* = \{c_1, \dots, c_k, c^*\}$, define $F^- = F^* \setminus \{c^*\}$. An item is retained only when F^* is satisfiable and target-determinate while F^- is satisfiable and target-ambiguous. Two source conditions share exactly F^- :

- **Omission:** the source retains the sentence supporting c^* ; grounded repair is correct.
- **Underspecified:** that sentence is removed; abstention is correct.

The solver state is identical within each pair, so ambiguity alone cannot reveal the label. The 120-pair confirmatory set is disjoint from the 80-pair pilot and 100-pair development sets and was frozen before outcome inspection. It contains rank-deficient systems (20), linear separation (20), moment problems (40), Chinese-remainder reconstruction (20), and graph path sums (20).

All four policies receive the same source, F^- , target, schema, and two-attempt budget. Self-review receives neither verdict nor grounding rule; grounded self-review adds the exact-span grounding rule but receives neither a solver verdict nor witnesses; determinacy-guided adds the typed ambiguity verdict without grounding; DeGro receives both the verdict and grounding rule. The main comparison does not expose witness values to the repair model. GPT-OSS 20B and 120B use medium reasoning effort and temperature 1.0, giving 960 complete records per model. The post-hoc Nemotron 3 Nano 30B addition uses the same frozen data, prompts, methods, temperature, and API `think=medium` request, yielding 960 complete records; this does not establish equal internal reasoning budgets. FDA is the primary outcome; RSR measures correct repair on omission cases, CAR measures correct abstention on underspecified cases, and $FDA = (RSR + CAR)/2$. Confidence intervals resample pairs; paired tests use exact two-sided McNemar tests with Holm correction across the two GPT-OSS tests. The later Nemotron test is exploratory and reported separately.

D Extended Controlled Results

The main paper reports all 300 MIRA-Math pairs (80 pilot, 100 development, 120 held-out), with 2,400 unique case-policy records per model. Pilot and development outcomes informed method development, so the full-set estimate is not an independent held-out test. The table below preserves the separate 120-pair results and their original analysis protocol for comparison. For GPT-OSS 20B, DeGro minus grounded self-review is +5.0 FDA points with pair-cluster 95% CI [0.4, 9.6] and exact

Table 1: Results on the fixed 120-pair set (percent). The 20B comparison was preregistered; the GPT-OSS 120B replacement and Nemotron Nano addition are post-hoc.

Model	Method	FDA	RSR	CAR	UR
GPT-OSS 20B	Self-review	90.4	93.3	87.5	15.2
	Grounded self-review	90.0	100.0	80.0	16.7
	Determinacy-guided	92.9	86.7	99.2	6.3
	DeGro	95.0	92.5	97.5	4.3
GPT-OSS 120B	Self-review	77.5	100.0	55.0	31.0
	Grounded self-review	87.5	95.8	79.2	17.9
	Determinacy-guided	91.3	98.3	84.2	15.1
	DeGro	93.8	97.5	90.0	10.7
Nemotron Nano 30B	Self-review	84.6	76.7	92.5	25.8
	Grounded self-review	89.6	80.8	98.3	3.0
	Determinacy-guided	73.8	50.8	96.7	24.7
	DeGro	90.8	81.7	100.0	1.0

$p = .050$. For GPT-OSS 120B, the difference is +6.25 points with CI [1.7, 10.8] and $p = .0135$. Holm correction across the two model-specific tests gives $p = .0501$ and $p = .0271$, respectively. The 120B substitution was selected after the original analysis plan and is treated as post-hoc.

For Nemotron 3 Nano 30B, DeGro changes FDA from 89.6% to 90.8% (+1.25 points, pair-cluster 95% CI [-2.5, 5.0], 12 versus 9 discordances, exploratory exact $p = .664$). This is statistically inconclusive. DeGro reaches 100.0% CAR versus 98.3% for grounded self-review, while UR falls from 3/100 proposed additions (3.0%) to 1/99 (1.0%). RSR is 98/120 versus 97/120. Verdict-only repair scores only 73.8% FDA.

Across all 300 pairs, GPT-OSS 20B changes from 91.2% to 96.2% FDA (+5.0, CI [2.3, 7.8], $p < .001$), while GPT-OSS 120B changes from 88.3% to 95.7% (+7.3, CI [4.7, 10.0], $p < .001$). This cumulative analysis is secondary because it includes

development observations. In a 100-pair exploratory ablation, exposing target-divergent witnesses to the repair model did not improve DeGro (98.5% versus 97.5% FDA; adjusted $p = .727$); witnesses remain valuable as audit evidence.

Across all 300 pairs, Nemotron Nano changes FDA from 89.5% to 90.3% (+0.83 points, pair-cluster 95% CI $[-1.5, 3.3]$; 30 versus 25 discordances; exact $p = .590$). The cumulative run has all 2,400 expected records: 640 pilot, 800 development, and 960 fixed-set records. DeGro reaches 100.0% CAR and 80.7% RSR, versus 99.0% and 80.0% for grounded self-review; UR is 2/244 versus 7/247 additions. The cumulative result remains statistically inconclusive. Transport retries retained the first valid response, without score-based resampling.

E Exploratory Source-Coverage Refinement

We inspected only the 80-pair development outputs before defining one candidate prompt. The refinement instructs the model to compare every stated condition algebraically against the current formalization, consider indirect relations among non-target variables, distinguish ambiguity of the current formalization from adequacy of the source, and treat listed domains as already enforced. Labels, gold targets, missing constraints, and pair identifiers are absent from the prompt. The candidate was frozen before any new-test inference.

The original 300 records exhaust the solver-compatible MIRA pool used by this study. We therefore generated a separate 100-pair synthetic stress set, with 25 pairs each for linear separation, graph path sums, Chinese-remainder reconstruction, and rank-deficient linear systems. All 200 gold actions pass the released scorer; all source strings are unique, and none exactly matches the original 300-pair source strings. Template similarity remains, so this set is not an external benchmark or an additional MIRA test. Both prompts were run concurrently with temperature 1.0 and API `think=medium`; inference retained the first valid response and never retried by score.

Table 2: Exploratory source-coverage refinement on 100 synthetic pairs (percent). UR is unsupported additions divided by all proposed additions.

Model	Prompt	RSR	CAR	FDA	ORR	UR
Nemotron Nano 30B	Original DeGro	52.0	99.0	75.5	1.0	1.9
	Source coverage	90.0	100.0	95.0	0.0	8.2
GPT-OSS 120B	Original DeGro	99.0	98.0	98.5	2.0	2.9
	Source coverage	100.0	99.0	99.5	1.0	1.0

For Nano, the FDA difference is +19.5 points (pair-cluster 95% CI $[13.5, 25.5]$; 45 discordances favor the refinement and six favor the original; exact $p = 1.83 \times 10^{-8}$). UR rises from 1/53 to 8/98 because the refined prompt attempts substantially more repairs. For GPT-OSS 120B, FDA changes by +1.0 point (CI $[-1.0, 3.0]$; three versus one discordances; $p = .625$), and UR falls from 3/102 to 1/101. Holm-adjusted p -values across these two model tests are 3.66×10^{-8} and .625. On the development set used to design the prompt, Nano FDA changes from 88.75% to 98.75%, and GPT-OSS 120B from 98.125% to 100%; these development values are descriptive. The synthetic test supports the mechanism for Nano but cannot establish generalization beyond these templates.

F Natural-Error Study

Exactly one Structured SymCode formalization was generated per problem and reused across methods. Cohort membership was frozen from parse, compiler, and solver states before annotation or repair-outcome inspection. Blinded annotation packets omitted the determinacy verdict, witness, repair, method, and answer outcome.

Two reviewers inspected all 292 MATH-500 and 503 ALG514 final AI-assisted labels. Because separate pre-discussion human labels were not retained, we claim two-person verification, not independent double annotation. Labels are CORRECT, MISSING-CONSTRAINT, WRONG-CONSTRAINT, EXTRA-CONSTRAINT, WRONG-DOMAIN, WRONG-TARGET, COMPUTATIONAL-ERROR, and OTHER; ambiguity itself never determines the label.

Table 3: Human-verified MATH-500 outcomes on 292 conclusive, solver-supported cases (58.4% coverage).

Label	<i>n</i>	Ambiguous	Repair	Solver correct	DeGro correct
Correct	158	33	10	119	126
Missing constraint	24	21	4	2	3
Wrong constraint	32	10	0	16	16
Extra constraint	8	4	1	4	4
Wrong domain	1	1	0	0	0
Wrong target	69	52	10	3	8
Total	292	121	25	144	157

Within the 24 target-critical omissions, ambiguity recall is 87.5% (Wilson 95% CI [69.0, 95.7]), detector precision is 17.4% (CI [11.6, 25.1]), and end-to-end grounded repair is 4.2% (CI [0.7, 20.2]). Overall accuracy changes from 49.3% to 53.8%; 16 discordances favor DeGro and three favor Structured Solver (exact McNemar $p = .00443$).

For ALG514, 503/514 problems are covered (97.86%). Labels comprise 477 correct, 11 wrong-target, eight target-critical missing-constraint, and seven wrong-constraint cases. Structured Solver scores 94.16% and DeGro 94.36% under the released scoring protocol; nine numerically compatible cases still carry a verified semantic error.

G Reproducibility Checklist

Table 4: Frozen artifacts and identifiers.

Artifact	Frozen value
MIRA source revision	72ff918901584d1d1e835a9f86d41e3d51a915a2
MIRA source SHA-256	78a021526f9457af6e3776ac470c4ef1202d18fd9a19539e7e289294997b2ad0
Selection seed	20270911
Confirmatory prompt SHA-256	ea-fed6be4973c21e1efbb5cc804112c9d50c95764e8f5066b58bdee58b92bdda
GPT-OSS 20B confirmatory SHA-256	5616d0c6203f762560439ee193f1148133d06baf034d6c34c7250a0a44ce3f0b
GPT-OSS 120B confirmatory SHA-256	38d18e6905fb3acafb236f9e6a4b0aa3aed2c2fb2dd8ece75f236074dbbf01067
Nemotron Nano 30B confirmatory SHA-256	59eaaa1b86fb26b333c0340448c833e7485e708e7f0b4442d04e8e91cb9ff595
Source-coverage synthetic set SHA-256	8ba85d5370803b2084dbf03cd2c8bcb438d5497a2efc47597320fda382837a7e
Source-coverage complete outputs SHA-256	b500c5b710a360f22a8697425439798ef66d95f4dabe89718fd4f2fcf9ae3131
MATH-500 verified labels	4e1bbe0b9eae-fdefe4569ec10fdd42daf8f1131d17dfbc410d0faed0b430ad38
ALG514 verified labels	72217047ff8d41b4a3766e504325821bd07230d7b6dc7636475294ab8e5ffecfcd

The released scripts reconstruct tables from JSONL records, preserve cohort provenance, and fail on missing expected counts. Statistical code implements pair-cluster bootstrap intervals, exact McNemar tests, Holm adjustment, and Wilson intervals. All 49 repository tests pass in an isolated environment. Exact prompts, schemas, and per-record logs accompany the code release rather than being reproduced here.

H AI-Assisted Technology Disclosure

OpenAI Codex (GPT-5 model family) assisted with prototype code, experimental scaffolding, language polishing, restructuring, and formatting. It is not an author; the human authors remain responsible for the methodology, implementation, experiments, analyses, citations, and claims. The pipeline image was supplied by the author, who should retain its provenance for venue disclosure if required.

I DRAW-Paired: Natural-Source Paired Evaluation

DRAW-Paired uses only the 600 training and 200 development problems of DRAW-1K [1]. Its 200 test problems are not used. One duplicate source identifier and 200 single-equation problems are excluded; the remaining 599 systems have a target-critical deletion verified by the solver. The initial seeded 300-pair cohort contains 223 training-derived and 77 development-derived

sources. A source-disjoint, seeded extension selects 50 of the remaining 67 unflagged proposals (37 training-derived and 13 development-derived), yielding an initial pool of 350 pairs and 700 cases. After a subsequent data re-review, the author requested exclusion of 50 pairs: 44 from the initial cohort and six from the extension. The retained benchmark has 300 pairs and 600 cases (221 training-derived, 79 development-derived), with 256 initial and 44 extension pairs. The exact identifiers are released in `excluded_pair_ids.txt`; no item-specific exclusion rationale was supplied. Each source contributes an omission case and an underspecified case with identical retained equations and target. Variables are real-valued and full-system target values are checked against published gold solutions. These are injected gold-equation omissions, not a natural model-error cohort.

The author confirmed manual review of the initial 300 proposed spans and deletions after provisional inference. Final validation detected two incorrectly segmented title abbreviations; both supporting sentences were corrected and all 16 case-policy records per model for those two pairs were rerun. Other responses were retained and rescored. The extension was selected after the initial outcomes and locked before its own inference. The author subsequently approved all 50 extension spans and deletions without changes; exact model-input equality was verified, so no additional inference was needed. The 300 retained pairs are author-reviewed and frozen. Both review and subsequent exclusions followed inference and access to outcomes; this subset analysis is post-hoc, not preregistered. No independent annotation or agreement statistic is claimed. Excluded outputs were deleted from released raw and complete result files. The retained initial cohort contributes 2,048 records and the retained extension contributes 352 per model; legacy filenames retain the original cohort sizes, while manifests report current counts. The main paper reports all four policies, not only the favorable comparison.

Each model has 2,400 complete unique records under temperature 1.0 and `API think=medium`. Primary tests compare DeGro with grounded self-review on FDA using exact McNemar, 10,000 pair-cluster bootstrap resamples, and Holm adjustment across three models. The reviewed initial dataset, approved extension packet, frozen retained 300-pair dataset, corrected raw responses, complete outputs, and analysis JSON retain separate provenance.

References

- [1] Shyam Upadhyay and Ming-Wei Chang. Annotating derivations: A new evaluation strategy and dataset for algebra word problems. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*, pages 494–504. Association for Computational Linguistics, 2017.